



Spring 2021–2022 (Math 102: Calculus II)

Second Mid-Term Exam: Suggested solutions

1. (5+5+5=15 Points)

(a) Let $a_n = 1 - n3^n$, for $n = 1, 2, \dots$. Since $(n+1)3^{n+1} > n3^n$, for any $n = 1, 2, \dots$, we conclude that $1 - (n+1)3^{n+1} < 1 - n3^n$ and hence $a_{n+1} < a_n$. Therefore, the sequence is decreasing.

Alternative solution: Consider the function $f(x) = 1 - x3^x$, for any $x \geq 1$. Since $f'(x) = -3^x(1 + x \ln 3) < 0$, for any $x \geq 1$, we deduce that f is decreasing and so is the sequence.

(b) We have $a_n = 1 - n3^n < 1$, for $n = 1, 2, \dots$ and hence the sequence is bounded above by 1.

Alternative solution: Since the sequence is decreasing, then $a_n \leq a_1 = 1 - 3 = -2$, for $n = 1, 2, \dots$ and hence the sequence is bounded above by -2 .

(c) The sequence cannot be bounded below since $\lim_{n \rightarrow \infty} a_n = -\infty$.

2. (15 Points) Knowing that the series converges when $x = 0$, assume that $x \neq 0$ and apply the ratio test for absolute convergence with $a_n = \frac{(-1)^n}{2^n \sqrt[3]{n}} x^n$. We have

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{(-1)^{n+1} x^{n+1}}{2^{n+1} \sqrt[3]{n+1}} \cdot \frac{2^n \sqrt[3]{n}}{(-1)^n x^n} \right| = \frac{|x|}{2} \lim_{n \rightarrow \infty} \sqrt[3]{\frac{n}{n+1}} = \frac{|x|}{2}.$$

Thus the series converges if $\frac{|x|}{2} < 1$, that is, $|x| < 2 = R$, or equivalently, $-2 < x < 2$.

At $x = -2$, the series becomes $\sum_{n=2}^{\infty} \frac{1}{\sqrt[3]{n}}$, which is a p -series with $p = 1/3 < 1$ and so diverges.

At $x = 2$, the series is: $\sum_{n=2}^{\infty} \frac{(-1)^n}{\sqrt[3]{n}}$. The latter converges by using the Alternating Series Test. Indeed, let

$b_n = \frac{1}{\sqrt[3]{n}} > 0$ for $n \geq 1$. We have: $\lim_{n \rightarrow \infty} b_n = \lim_{n \rightarrow \infty} \frac{1}{\sqrt[3]{n}} = 0$. Moreover, $b_{n+1} = \frac{1}{\sqrt[3]{n+1}} < b_n = \frac{1}{\sqrt[3]{n}}$.

The conditions of the alternating series test are satisfied and the series must converge. Thus the interval of convergence is $(-2, 2]$ and the radius of convergence is $R = 2$.

3. (15 Points) We know that $\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}$, for all x and hence

$$\frac{\sin(x) - x}{x} = \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n},$$

whose radius of convergence is $R = \infty$.

Integrating both sides of the last identity and thanks to Theorem 2, p. 754, one can write:

$$\int \frac{\sin(x) - x}{x} dx = \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n+1)!} \int x^{2n} dx = \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n+1)(2n+1)!} x^{2n+1} + C, \quad C \in \mathbb{R},$$

with radius of convergence $R = \infty$.

4. **(7+8+10=25 Points)**

(a) This is a geometric series with $r = -\frac{1}{\pi}$ and hence the series converges as $|r| = \left| -\frac{1}{\pi} \right| < 1$. Moreover, the sum is $\frac{3}{1 + \frac{1}{\pi}} = \frac{3\pi}{1 + \pi}$.

(b) It suffices to write the series in the form $-\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} (-1/4)^n$, which matches the Maclaurin series of $\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} x^n$ ($|x| < 1$), with $x = -1/4 \in (-1, 1)$ and hence the given series converges and its sum is $-\ln(1 - (1/4)) = \ln(4/3)$.

(c) Since $\ln\left(\frac{2+n}{1+n}\right) = \ln(2+n) - \ln(1+n)$, we obtain the partial sum of the series

$$s_n = \ln(2) - \ln(1) + \ln(3) - \ln(2) + \cdots + \ln(n) - \ln(n-1) + \ln(n+1) - \ln(n) = \ln(n+1).$$

Consequently, $\lim_{n \rightarrow \infty} s_n = \infty$ and hence the series diverges.

5. **(10 Points)** Let $a_n = \frac{(n!)^n}{n^{3n}}$. Then, We have:

$$\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \lim_{n \rightarrow \infty} \frac{n!}{n^3} = \lim_{n \rightarrow \infty} \frac{n(n-1)(n-2)}{n \cdot n \cdot n} (n-3)! = \infty.$$

Hence the Root Test implies the divergence of the series.

6. **(5+5=10 Points)**

(a) As $\sum_{n=0}^{\infty} c_n x^n$ diverges when $x = 3$, it implies that this series diverges for at least $x \geq 3$ and $x < -3$ (see Theorem 4 p. 749). Therefore, $\sum_{n=0}^{\infty} c_n 4^n$ diverges.

(b) Since $\sum_{n=0}^{\infty} c_n x^n$ converges when $x = -1$, we conclude that it converges for at least x in $[-1, 1)$ (see Theorem 4 p. 749). Thus, the series $\sum_{n=0}^{\infty} c_n (1/2)^n$ converges.

7. **(5+5=10 Points)**

(a) We notice that $\lim_{n \rightarrow \infty} n \left(e^{\frac{1}{n}} - 1 \right)$ has the indeterminate form $0 \cdot \infty$. Letting $f(x) = x \left(e^{\frac{1}{x}} - 1 \right) = \frac{e^{\frac{1}{x}} - 1}{1/x}$, we have $\lim_{x \rightarrow \infty} f(x)$ has the form $\frac{0}{0}$. Hence, we can use l'Hospital's Rule to obtain

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{e^{\frac{1}{x}}(-1/x^2)}{(-1/x^2)} = e^0 = 1.$$

Thus, $\lim_{n \rightarrow \infty} n \left(e^{\frac{1}{n}} - 1 \right) = 1$.

(b) Let us use the Limit Comparison Test with $a_n = e^{\frac{1}{n}} - 1 > 0$ and $b_n = 1/n > 0$. We have:

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} n \left(e^{\frac{1}{n}} - 1 \right) = 1 > 0, \quad \text{by using (a).}$$

Since $\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n}$ is the harmonic series and hence diverges, so does $\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} \left(e^{\frac{1}{n}} - 1 \right)$.